

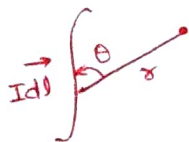
MOVING CHARGES & MAGNETISM

Biot-Savart's Law:- Magnetic field due to small current element.

$$d\vec{B} = \left(\frac{\mu_0}{4\pi}\right) \frac{Id\vec{l} \times \vec{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{Id\vec{l} \times \hat{r}}{r^2}$$

In vector form

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin\theta}{r^2}$$



• S.I Unit = Tesla (T)

• C.G.S Unit = Gauss $1T = 10^4 G$

• Dimension:- $F = qvB$, $B = \frac{F}{qv} = \frac{MLT^{-2}}{qV} = \frac{MLT^{-2}}{AT LT^{-1}}$
 $B = MT^{-2}A^{-1}$

• $\left(\frac{\mu_0}{4\pi}\right) = 10^{-7} Tm/A$ $\mu_r = \frac{\mu_m}{\mu_0}$ = Relative permeability
 ↳ Unitless & dimensionless

Similarities and differences b/w $\vec{B}$ & $\vec{E}$

- $\vec{B}$ due to vector source, $\vec{E}$ due to scalar.
- $\vec{B}$ and $\vec{E}$ both linear inside source.
- Both are long range, inverse square law.
- $\vec{B}$ is to force & position vector from source, $\vec{E}$ along the force & position vec.
- The principle of superposition is applied to both fields.
- There is angle dependence in the Biot-Savart's law which is not present in the electrostatic field. Magnetic field at the point in the dirⁿ of $d\vec{l}$ is zero.

MR* To find dirⁿ of Magnetic field

$$d\vec{B} = d\vec{l} \times \vec{r}$$

(Result) (1st vector) (2nd vector)

↳ Place your four-finger (Palm) of right hand along 1st vector slap 2nd vector, thumb will represent $\vec{B}$

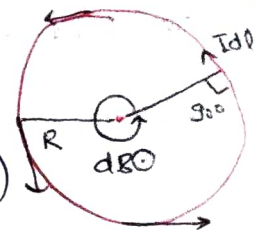
Magnetic field on the centre of Circular loop:-

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin 90^\circ}{R^2} = \frac{\mu_0}{4\pi} \frac{Idl}{R^2}$$

$$B = \frac{\mu_0}{4\pi} \frac{I 2\pi R}{R^2}$$

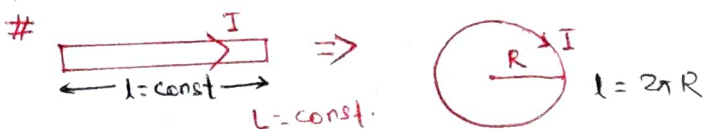
$$B = \frac{\mu_0 I}{2R} \quad \theta_{rad} = B = \frac{\mu_0 I}{2R} \left(\frac{\theta}{2\pi}\right)$$

Anticlock $\odot$, clockwise $\otimes$



$$B = n \left[\frac{\mu_0 I}{2R} \right]$$

↳ no of loop
 $R = \text{constant}$



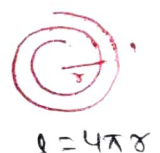
↳ straight wire bend into 1 circular loop

$$B_0 = \frac{\mu_0 I}{2R} = \frac{\mu_0 I}{2 \frac{l}{2\pi}} = \frac{\mu_0 I \pi}{l} \quad \because l = 2\pi R$$

↳ straight wire bend into 2 circular loop

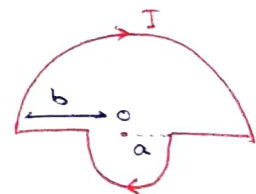
$$B = 2 \left[\frac{\mu_0 I}{2R} \right] = \frac{\mu_0 I}{R} 4\pi = 4B_0$$

* $B \propto n^2$



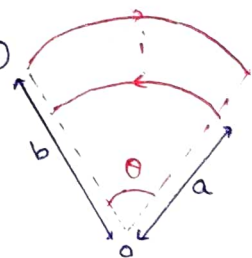
$$B_0 = \frac{\mu_0 I}{4b} \otimes + \frac{\mu_0 I}{4a} \otimes$$

$$\therefore B_0 = \frac{\mu_0 I}{2R} \left(\frac{\pi}{2\pi}\right)$$



$$B_0 = \frac{\mu_0 I}{2a} \left(\frac{\theta}{2\pi}\right) \odot - \frac{\mu_0 I}{2b} \left(\frac{\theta}{2\pi}\right) \otimes$$

$$B_0 = \frac{\mu_0 I \theta}{4\pi ab} (b-a)$$



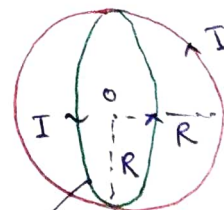
$$B_{\odot} = \frac{\mu_0 I}{2R}$$

$$B_{\text{centre}} = \sqrt{2} \left(\frac{\mu_0 I}{2R} \right)$$

$$B_{\text{centre}} = \sqrt{3} \left(\frac{\mu_0 I}{2R} \right)$$

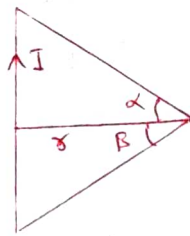
(I, R) → I is to the plane of paper

↳ If 3 circle placed concentric & plane is to each other.



Mag. field due to straight wire

$$B = \frac{\mu_0 I}{4\pi r} (\sin\alpha + \sin\beta)$$



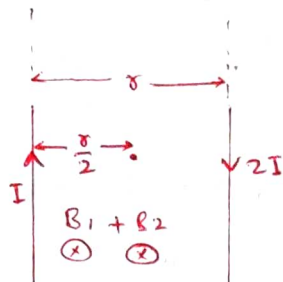
Infinite $B = \frac{\mu_0 I}{2\pi r}$

Semi-infinite $B = \frac{\mu_0 I}{4\pi r}$

$B = B_1 + B_2$

$$= \frac{\mu_0 I}{2\pi \frac{r}{2}} + \frac{\mu_0 2I}{2\pi (\frac{r}{2})} \otimes$$

$$= \frac{3\mu_0 I}{\pi r} \otimes$$

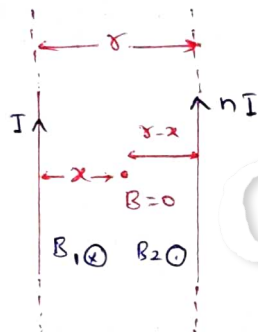


Find position where Mag. field is zero

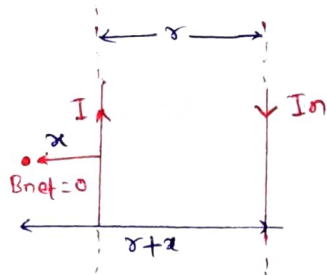
$x = \frac{r}{n+1}$

If $n=1$

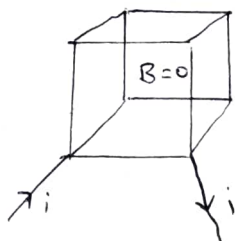
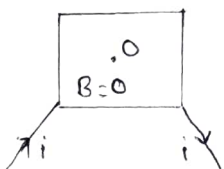
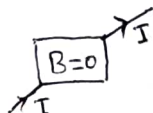
$$x = \frac{r}{2}$$



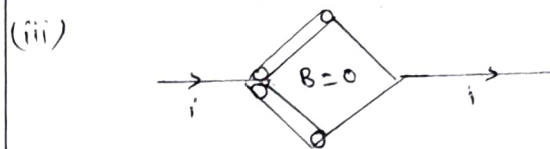
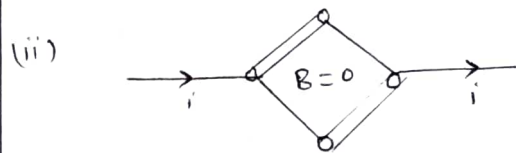
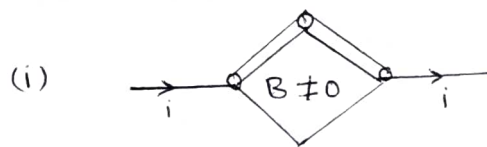
$x = \frac{r}{n-1}$



Symmetrical object

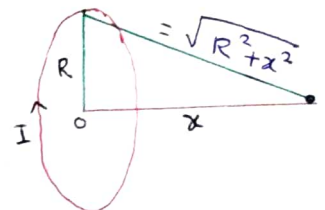


Combination of two thick & two thin wire :-



Mag. field on the axis of circular loop

$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$$



If $x=0$

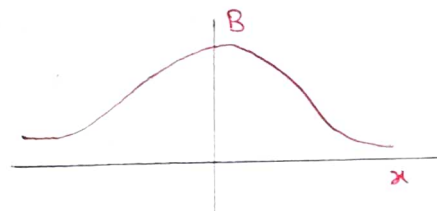
$$B = \frac{\mu_0 I}{2R}$$

If $x \ll R$

$$B = \frac{\mu_0 I}{2R}$$

If $x \gg R$

$$B = \frac{\mu_0 I R^2}{2x^3} \propto \frac{1}{x^3}$$



Coloumbe law

Biot Savart law

Gauss law

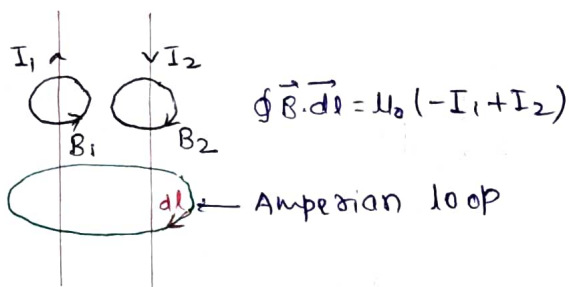
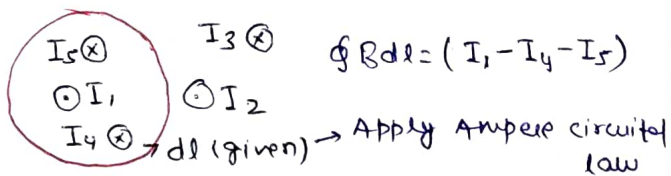
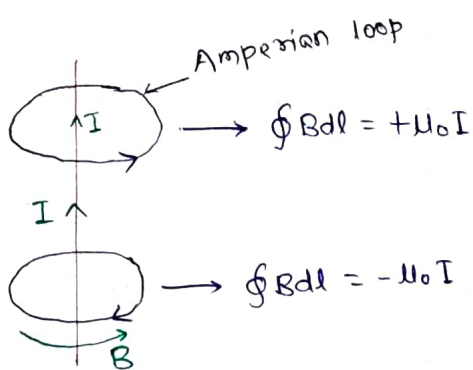
Ampere circuital law

Amperes Circuital law

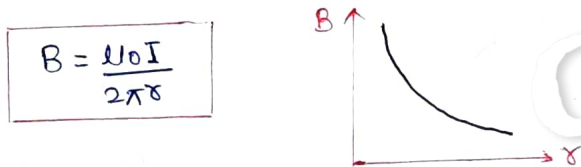
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{in}$$

↳ Enclosed current

- $\vec{B}$ due to all inside or outside current
- does not give magnetic flux
- Always valid & Applicable.
- Not Applicable or useful to calculate magnetic field.
- Amperian loop must be passing through point where $\vec{B}$ have to calculate.



Mag. field due to straight infinite current carrying wire:-



M.f due to Infinite hollow cylinder

At inside ($r < R$)

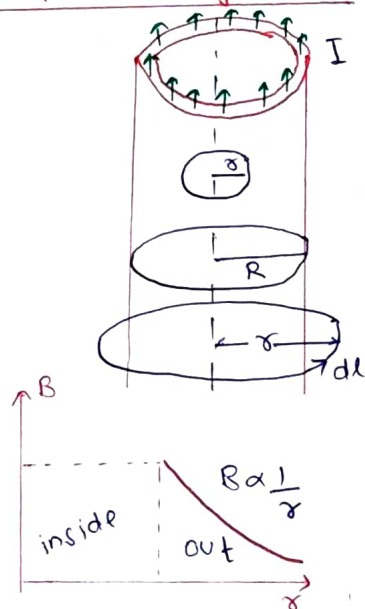
$B_{in} = 0$

At surface ($r = R$)

$B = \frac{\mu_0 I}{2\pi R}$

At outside ($r > R$)

$B = \frac{\mu_0 I}{2\pi r}$

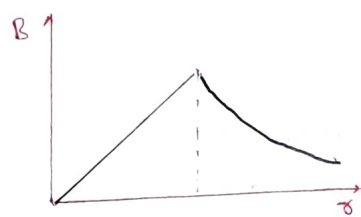
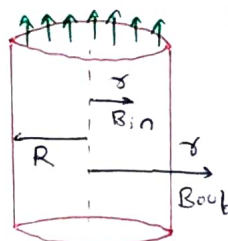


M.f due to Infinite hollow cylinder

$B_{surf} = \frac{\mu_0 I}{2\pi R}$

$B_{out} = \frac{\mu_0 I}{2\pi r}$

$B_{in} = \frac{\mu_0 I r}{2\pi R^2} = \frac{\mu_0 J r}{2}$ [$\because J = \frac{I}{\pi R^2}$]



M.f due to infinite current carrying solid cylinder where $I = J_0 \pi r^2$. Find M.f outside the cylinder:-

$I = \frac{J_0 2\pi R^3}{3}$

$B = \frac{\mu_0 J_0 R^3}{3r}$

To calculate M.f, steps to apply Ampere's Circuital law:-

- 1 Draw close symmetric amperian loop that must be passes through the point where field have to calculate
- 2 Find current enclosed by the amperian loop.
- 3 Calculate the integral $\oint \vec{B} \cdot d\vec{l}$ around the closed loop.
- 4 Equate $\oint \vec{B} \cdot d\vec{l}$ with $\mu_0 I$ & solve $\vec{B}$.
- 5 Angle b/w $\vec{B}$ & $d\vec{l}$ must be $0^\circ, 90^\circ$ or 180° .
- 6 Value of $\vec{B}$ must be same at all point.

• Infinite Solenoid



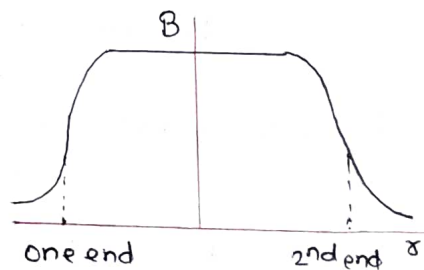
$B = \mu_0 n I = \frac{\mu_0 N I}{L}$

N = total turn

L = Total length

$n \Rightarrow$ turns per unit length

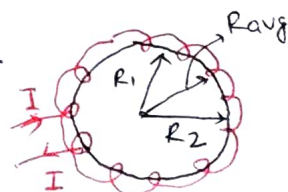
$n = \frac{N}{L}$



• Toroid $\rightarrow$ Circular Solenoid

$B = \mu_0 n I = \frac{\mu_0 N I}{L} = \frac{\mu_0 N I}{2\pi R_{avg}}$

$R_{avg} = \frac{R_1 + R_2}{2}$



Magnetic force

$$\vec{F} = q(\vec{v} \times \vec{B}) = qvB \sin \theta$$

θ = angle b/w $\vec{v}$ & $\vec{B}$.

- $\vec{F} \perp \vec{v}$ & $\vec{B}$ ($\vec{B} \perp \vec{a}$) ($\vec{a} \perp \vec{v}$)
- Speed = always const, momentum is variable bcz of direction.
- $K.E = \text{const}$, work = $\Delta K.E = \text{zero}$, power = 0
- If $\vec{u}$ is parallel or anti-parallel to magnetic field then force will be zero.
- If charge is at rest the magnetic force will be zero.
- Mag. field can only change dirⁿ of motion of charge particle.
- $\vec{F} \cdot \vec{B} = 0$, $\vec{F} \cdot \vec{u} = 0$, $\vec{a} \cdot \vec{u} = 0$

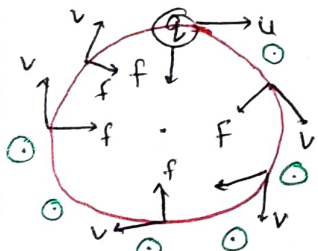
Path of charge particle in Mag. field

(i) charge is projected $\parallel$ to $\vec{B}$

$$\begin{aligned} \text{Force} &= 0 \\ a &= 0 \\ \vec{v} &= \text{const.} \end{aligned}$$

Magnetic force $\Rightarrow$ centripetal force

(ii) charge is projected $\perp$ to Mag. field



$$qvB \sin 90^\circ = \frac{mv^2}{R}$$

$$R = \frac{mv}{qB}$$

$R \propto m, R \propto \frac{1}{q}$
 $R \propto v, R \propto \frac{1}{B}$

$$F = qvB$$

$$F_c = qvB = \frac{mv^2}{R} \text{ (Inst)}$$

$(F_{\text{Avg}})_{\text{magnetic}} = \frac{qvB \sin(\theta/2)}{(\theta/2)}$

Radius of circular path

$$R = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mkE}}{qB} = \frac{\sqrt{2mqv}}{qB}$$

Time period

$\Rightarrow$ Time to complete one rotation (2π)

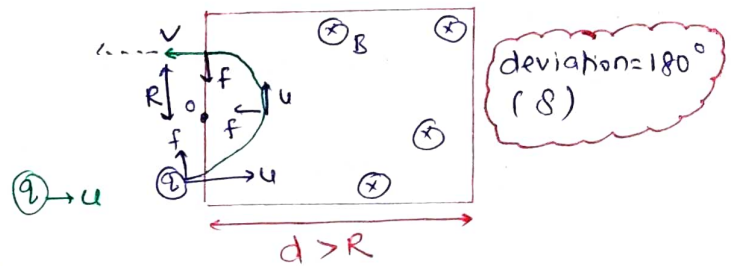
$$T = \frac{2\pi m}{qB} \rightarrow \text{Time doesn't depend upon speed.}$$

$\Rightarrow$ Time to complete ' θ ' Angle

$$T = \frac{m\theta}{qB}$$

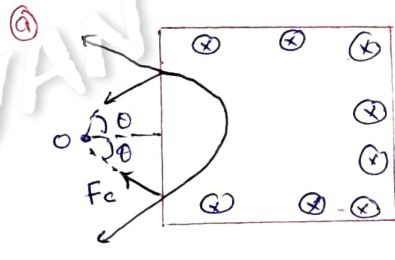
Frequency $\Rightarrow f = \frac{qB}{2\pi m} = \frac{1}{T}$

Charge is projected $\perp$ to Mag. field, but from outside $\&$ region of Mag. field.



Angle b/w Boundary & velocity is 90°

(a)

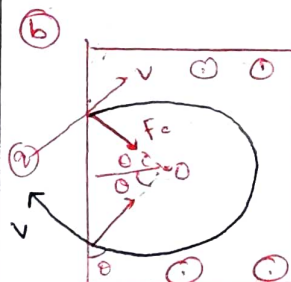


Total deviation = 2θ

$$T_\theta = \frac{m(2\theta)}{qB}$$

$$F_{\text{avg}} = \frac{qvB \sin \theta}{\theta}$$

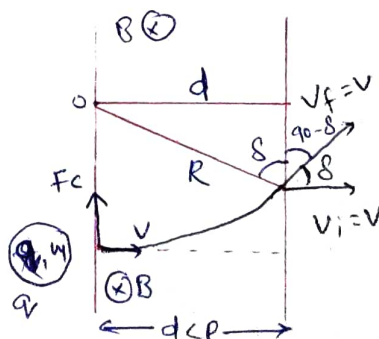
(b)



deviation = $(2\pi - 2\theta)$ (θ)

$$\text{Time} = \frac{m\theta}{qB}$$

Charge particle is projected outside from a region of mag. field but ($d < R$) then find deviation?

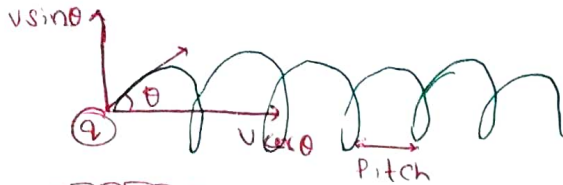


$\sin \theta = \frac{d}{R}$

$$\theta = \sin^{-1}\left(\frac{d}{R}\right)$$

Particle is projected at an angle ($\theta < 90^\circ$) from mag. field.

→ Path of the particle will be helical.

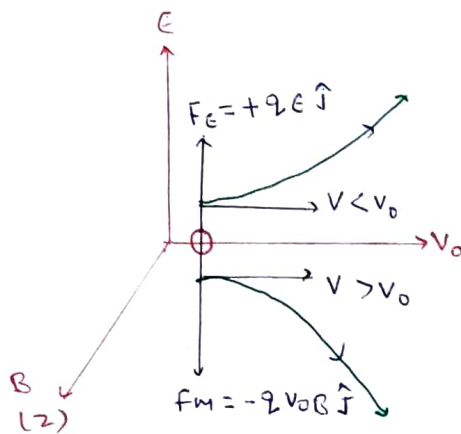


$$R = \frac{mv \sin \theta}{qB}$$

$$T = \frac{2\pi m}{qB}$$

$$\text{Pitch}(x) = v_x T = v \cos \theta \cdot \frac{2\pi m}{qB}$$

Velocity Selector



$$I \mathbf{B} \cdot \mathbf{E} = f_M$$

$$qE = qv_0 B$$

$$v_0 = \left[\frac{E}{B} \right]$$

Statement

rest ($v=0$)
 ⇒ (q) → we can't say about presence of mag. field.
 → we can say about electric field.

S-1 → Charge particle is at rest, experience zero electromagnetic force, the mag. field may be zero.

S-2 → Charge particle is at rest, experience electromagnetic force, then electric field must be non-zero.

S-3 → Charge is moving in E-M field without deviation, then electric field & Mag. field both may be non-zero or both may be zero.

S-4 → Charge particle passes E-M field without acceleration then '3' possible condition.

$$(E=0, B=0), (E=0, B \neq 0), (E \neq 0, B \neq 0) \quad (u \parallel B)$$

S-5 → charge is moving on circular path in E-M field then only possible condn, ($E=0, B \neq 0$)

Magnetic force on a current carrying wire

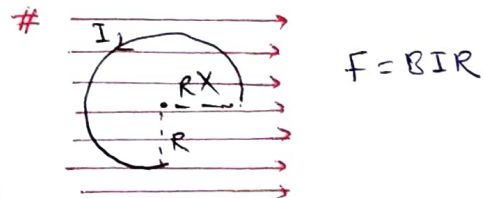
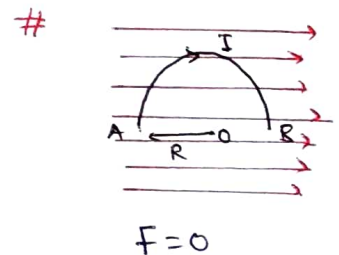
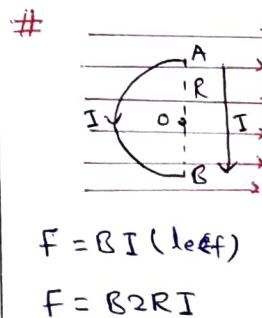
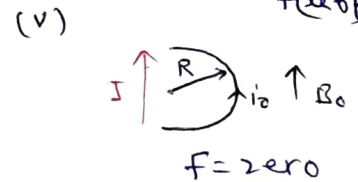
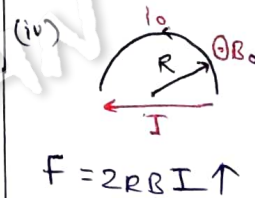
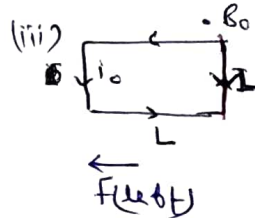
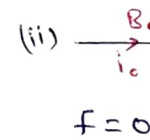
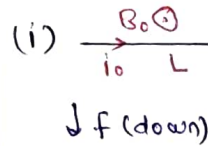
$$\mathbf{F} = \mathbf{I} \mathbf{L} \times \mathbf{B} = ILB \sin \theta$$

→ Force must be $\perp$ to $\mathbf{B}$ & $\mathbf{I}$.

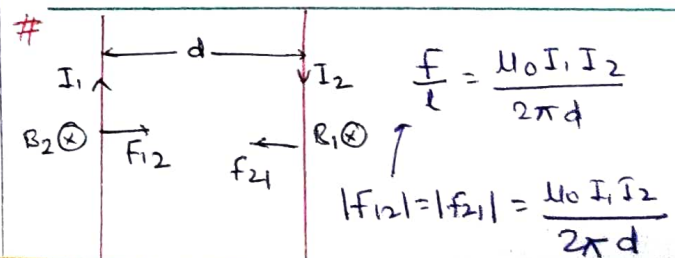
- Direction of force can be determined by using Fleming's left hand rule.
- Force on a closed loop (of any shape) placed in a uniform magnetic field is zero, as $\mathbf{I}_{\text{eff}} = 0$.



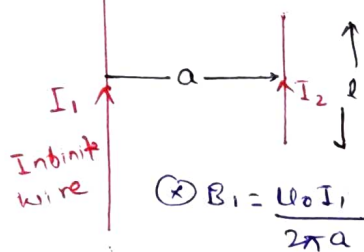
Q → Magnetic force on different wire



Force on small current carrying wire due to Infinite large current carrying wire :-



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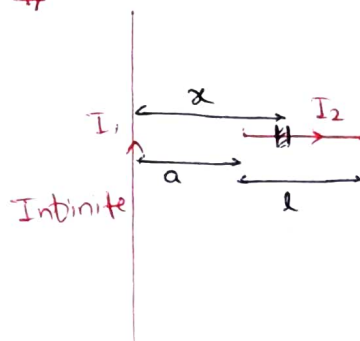


$$\otimes B_1 = \frac{\mu_0 I_1}{2\pi a}$$

$$F_{21} = B_1 I_2 l$$

$$F_{21} = \frac{\mu_0 I_1 I_2 l}{2\pi a}$$

#



$$F = \frac{\mu_0 I_1 I_2 l \log\left(\frac{a+l}{a}\right)}{2\pi}$$

Biot-Savart law for moving charge

$$B = \frac{\mu_0}{4\pi} \frac{qV \sin\theta}{r^2}$$

$$\frac{F_{mf}}{F_{ef}} = \frac{v^2}{c^2} \left[c^2 = \frac{1}{\mu_0 \epsilon_0} \right]$$

$$F_{ef} \gg F_{mf}$$

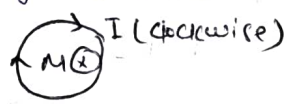
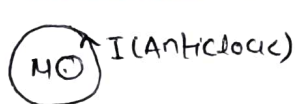
Circular current loop as a magnetic Dipole:-

$$B = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}$$

$$B = \frac{2\mu_0}{4\pi} \times \frac{R^2 \pi i}{x^3} = \frac{2\mu_0 i}{x^3} = \frac{2\mu_0 i A}{x^3}$$

$$\Rightarrow \vec{M} = I \vec{A} \rightarrow \text{Vector}$$

$\rightarrow$ dirn along Mag. field



$$\text{no. of loop:- } M = NIA = N I \pi R^2$$

Torque on a current loop (Magnetic dip)

$$\vec{\tau} = \vec{M} \times \vec{B} = MB \sin\theta \rightarrow \text{Angle b/w } \vec{M} \text{ \& } \vec{B}$$

$$\vec{\tau} = MB \cos\theta \rightarrow \text{Angle b/w plane of loop and } \vec{B}$$

$$\tau = BIA \bullet \text{Net force} = \text{zero}$$

$$\bullet \vec{\tau} \perp \vec{M} \text{ \& } \vec{B}$$

Magnetic potential Energy stored in Magnetic dipole:-

$$U = -MB \sin\theta$$

$$= -\Delta U = U_i - U_f$$

$$\text{work done, } W_{cf} = +\Delta U = U_f - U_i$$

Time period:-

$$T = 2\pi \sqrt{\frac{I}{MB}}$$

Gyromagnetic ratio:-

$$\frac{M}{L} = \frac{e}{2m_e} = \frac{8.8 \times 10^{-10} \text{ C}}{\text{kg}}$$

Bohr's magneton:-

$$\mu_B = \frac{e}{2m_e} \left[\frac{h}{2\pi} \right]$$

Moving coil Galvanometer

$\Rightarrow$ Working principle:- Torque on a magnetic dipole in uniform magnetic field.

$$\tau = MB = N I A B = C \theta$$

$$\theta = \frac{B I N A}{C}$$

$$C_{spring} = C \theta$$

$\rightarrow$ torsional const?

$$I_s = \left(\frac{\theta}{I} \right) = \frac{NAB}{C} \rightarrow \text{Current Sensitivity}$$

Divided by 'R' both sides

$$V_s = \frac{I_s}{R} = \frac{NAB}{CR} = \frac{\theta}{V} \rightarrow \text{Voltage Sensitivity}$$